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Deep Learning for Art Classification

Authors: Andreea Roica, Jenny Cubelo, Libero Biagi, Marisa Esteves, Oliver Kain
Student IDs: 20250361, 20250431, 20250349, 20250348, 20250401
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University: NOVA Information Management School
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1 Introduction

The classification of artistic images poses unique challenges compared to conventional image recognition tasks. Paintings often exhibit significant variability in style, composition, and abstraction, even within the same artist class, while distinct classes may share similar visual characteristics. As a result, effective models must capture subtle patterns such as texture, color distribution, and stylistic features that are commonly learned through deep convolutional representations [2, 1].

This project addresses the problem of classifying artworks from a subset of the WikiArt dataset using deep learning techniques. The objective is to develop models capable of learning discriminative visual representations that generalize well despite the complexity and variability of the data.

To this end, several deep learning approaches are explored and evaluated, with a focus on understanding how different architectures capture the underlying structure of artistic images. Convolutional neural networks have been shown to be effective at learning hierarchical feature representations from images, while also being applicable to style- and texture-sensitive visual tasks [1]. The performance of these models is assessed using standard evaluation metrics, providing insight into their effectiveness for this task.

2 Dataset and Data Preparation

2.1 Dataset Description

The dataset used in this project is a subset of the WikiArt collection, which contains digitized paintings from 23 artists. The task considered is artist classification, where each image is associated with the artist who created the work, forming a multi-class classification problem. The distribution of samples is not perfectly balanced across classes, with the number of images per artist ranging from 336 to 1322, for a total of 13,340 images. This imbalance introduces additional challenges for model training and evaluation.

2.2 Data Splitting Strategy

To ensure that the model generalizes across different visual styles of each artist and to address the class imbalance, we adopted a clustering-based splitting strategy within each artist. For each image, we extracted low-level visual features, including a grayscale indicator and statistical descriptors in HSV color space, after resizing images to 64×64 and normalizing pixel values. Clustering was performed independently per artist using KMeans, with the number of clusters selected by maximizing the silhouette score. This groups artworks with similar visual characteristics (e.g., color palette or style) within each artist.

Initially, we experimented with a purely random split by artist, but observed imbalanced results. This led to situations where the model was not exposed during training to certain stylistic variations, negatively affecting performance.

The final dataset split was obtained by stratifying on the combined artist–cluster label, using an 80/10/10 ratio for training, validation, and testing. This ensures that each split contains samples from all clusters of every artist, exposing the model to the full range of styles during training. The 80/10/10 ratio was chosen to maximize the number of training samples while maintaining sufficiently large and balanced validation and test sets.

Stratification by artist–cluster helps preserve both class distribution and intra-class variability across splits, leading to a more robust and fair evaluation of the classification model.

2.3 Data Preprocessing

2.3.1 Normalization

For the purpose of normalization, images were decoded into RGB format and pixel values were scaled from the range [0,255] to [0,1], ensuring consistent input representation. The same preprocessing was applied uniformly to the training, validation, and test sets. only in our net. the other models were pre trained and we used the normalization that they were trained with.

2.3.2 Resizing

We analyzed the resolution of all images in the dataset by extracting their dimensions. The results showed that all images share a resolution of 512×512 pixels, which is relatively high for efficient model training. Therefore, images were considered for downscaling to a smaller size to reduce computational cost.

2.4 Data Augmentation

Techniques used (e.g., rotations, flips, color jitter) and rationale.

2.5 Batching

The dataset was processed in batches of 64 samples, as it provides a good balance between memory usage and training performance. Each batch consists of tensors of shape (64, 512, 512, 3) for images and (64,) for labels.

3 Methodology

3.1 Custom CNN Architecture

The proposed CNN follows a design in which the number of filters increases progressively across convolutional blocks (32, 64, 128, 256). This allows the network to learn increasingly complex representations: early layers capture low-level features such as edges and colors, while deeper layers encode higher-level patterns such as textures and stylistic characteristics. The increase in filter depth also compensates for the reduction in spatial resolution caused by pooling operations.

All convolutional layers use 3×3 kernels with `same` padding, preserving spatial dimensions and allowing the network to keep extracting features at the same resolution before downsampling is applied later via pooling.

Instead of using a flattening operation, we apply Global Average Pooling at the end of the convolutional stack. This reduces each feature map to a single value, drastically lowering the number of parameters.

The final classifier consists of two fully connected layers with dropout regularization, followed by a softmax output layer for multi-class prediction.

This architecture was achieved by trial and error until performance plateaued at a weighted F1-score of 0.58. Furthermore, we reached the conclusion that our model used mainly color as a discriminative feature (likely influenced by the limited number of training samples). As a result, we decided to adopt transfer learning with pre-trained models in order to capture more complex features such as shapes, textures, and corners.

3.2 Transfer Learning with Pre-trained Models

For all architectures described below, pretrained weights from ImageNet were used and the backbone networks were treated as fixed feature extractors, with their layers frozen during training. A custom classification head was added on top to adapt each model to the artist classification task.

3.2.1 ResNet50

ResNet50 is a deep convolutional neural network architecture composed of 50 layers, based on the concept of residual learning. Its key feature is the use of residual connections, which allow the network to learn residual mappings instead of direct transformations. These connections facilitate the learning of both low-level and high-level features, improving the model's capacity to generalize across varying artistic styles and making ResNet50 a suitable choice for our problem.

3.2.2 InceptionV3

InceptionV3 is a convolutional neural network architecture designed to capture visual features at multiple scales. Its main idea is the use of Inception modules, where multiple operations (e.g., 1×1, 3×3, 5×5 convolutions and pooling) are applied in parallel to the same input and their outputs are concatenated. This allows the model to extract different types of features simultaneously. This architecture is particularly useful when the size of relevant patterns is unknown, as it can capture both fine details and more global structures. For our task, this is beneficial since artistic styles can vary in both local features and overall composition.

3.2.3 Vision Transformer (ViT)

ViT is an architecture that applies transformer models to image classification. Instead of using convolutions, the image is divided into fixed-size patches, which are treated as a sequence and processed using self-attention mechanisms, allowing the model to capture global relationships between different parts of the image. This is useful for artist classification, where overall composition and spatial relationships can be important.

4 Experimental Setup

All models were trained using the Adam optimizer with a learning rate of 10^{-3} and the sparse categorical cross-entropy loss function, which is suitable for multi-class classification with integer-encoded labels.

Model performance was primarily evaluated using the F1-score, as it provides a better assessment under class imbalance, although accuracy was also monitored during training.

Training was performed for a maximum of 60 epochs. To prevent overfitting and reduce unnecessary computation, early stopping was applied based on the validation loss, with a patience of 10 epochs for CNN-based models and 5 epochs for the Vision Transformer. The best model weights were restored after training.

Additionally, a learning rate scheduler (ReduceLROnPlateau) was used to reduce the learning rate by a factor of 0.5 when validation loss plateaued, with a minimum learning rate of 10^{-6}. Model checkpoints were saved based on the best validation loss.

5 Results and Evaluation

5.1 Model Comparison

Table 1 presents the performance of all evaluated models.

Table 1: Model Comparison

Model	F1-Weighted	Val Acc	Val Loss	Train Acc	Train Loss	Avg Epoch Time (s)
Our Net	0.54	0.54	1.52	0.58	1.32	56.91
ResNet50	0.79	0.79	0.79	0.97	0.10	63.98
InceptionV3	0.69	0.70	1.00	0.87	0.44	69.33
ViT	0.28	0.34	2.35	0.34	2.26	192.49

ResNet50 achieved the best overall performance across all metrics, with the highest F1-score and validation accuracy, while maintaining a reasonable computational cost.

InceptionV3 showed competitive but lower performance, whereas the custom CNN performed significantly worse, highlighting the benefits of transfer learning in capturing complex artistic features.

The ViT performed substantially worse than all other models, with lower accuracy and higher loss, and also required significantly longer training time.

For these reasons, ResNet50 was selected as the baseline model for further analysis.

5.2 Training Behavior of ResNet50

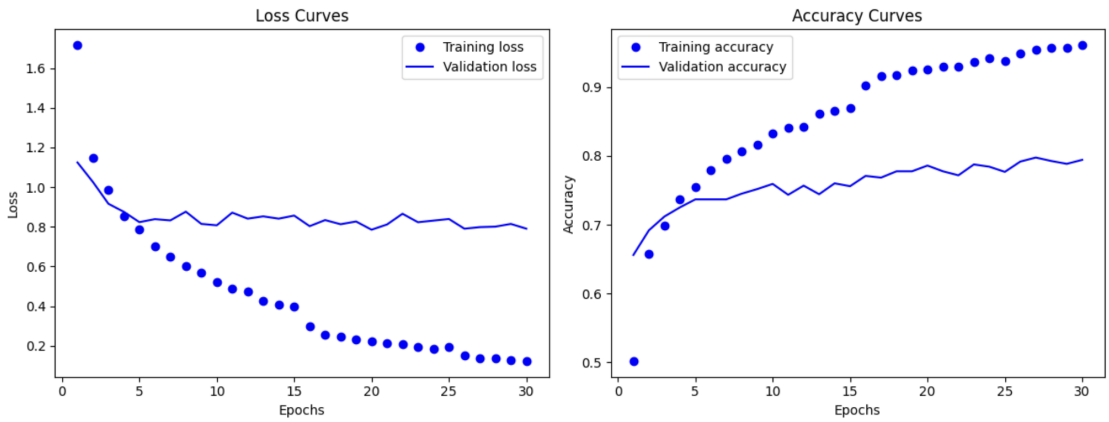

Figure 1: ResNet50 Training and Validation Curves

Figure 1 shows the training and validation loss and accuracy curves for ResNet50. The model exhibits a steady decrease in training loss and a corresponding increase in training accuracy, indicating effective learning. Validation performance improves rapidly during the initial epochs and stabilizes after. A noticeable gap between training and validation curves emerges as training progresses, with training accuracy reaching approximately 97% while validation accuracy plateaus around 79%. Similarly, training loss continues to decrease, whereas validation loss stabilizes around 0.79, which indicates moderate overfitting. Despite this, the validation performance remains strong, suggesting that the model generalizes well overall. The use of early stopping and learning rate scheduling helps prevent further overfitting and ensures stable convergence.

5.3 Per-Class Performance of ResNet50

Table 2: Classification report per class of ResNet50

Class	Precision	Recall	F1-score	Support
7	0.96	1.00	0.98	47
14	0.90	0.90	0.90	115
0	0.89	0.90	0.90	52
17	0.93	0.87	0.90	87
9	0.80	0.95	0.86	37
...	...	...	...	...
15	0.46	0.67	0.54	48
21	0.56	0.47	0.51	30

A per-class analysis of ResNet50 reveals variability in performance across artists. Several classes achieve high F1-scores, indicating that the model effectively captures distinctive stylistic features. In contrast, lower performance is observed for certain classes (e.g., classes 15 and 21), where F1-scores drop below 0.6. This variation can be attributed to two main factors. First, class imbalance affects performance, as classes with fewer samples tend to be harder to learn. Second, some artists exhibit similar visual styles, making them more difficult to distinguish. Overall, while the model performs well on average, these results highlight the challenges of intra-class variability and inter-class similarity in artist classification.

6 Fine-tuning of ResNet50

7 Discussion

The strong performance of ResNet50 is likely due to the effectiveness of residual connections and pretrained convolutional features, which allow the model to learn rich visual representations even with a relatively limited dataset. ADD ABOUT FINE-TUNING.

InceptionV3 also benefits from transfer learning and multi-scale feature extraction, but its more complex architecture may make optimization slightly less stable in this setting.

The custom CNN underperforms compared to pretrained models, which is expected given that it must learn all feature representations from scratch without large-scale pretraining.

The ViT performs worst, which can be attributed to its reliance on large-scale data and lack of built-in inductive biases, i.e., it does not inherently exploit local spatial patterns as convolutional models do. With a smaller dataset and without fine-tuning, it struggles to learn effective representations.

Overall, the results suggest that transfer learning with convolutional architectures is better suited for this task than training from scratch or using transformer-based models under limited data conditions.

8 Conclusion

Summary of findings, possible improvements, future work.

References

[1] Leon A. Gatys, Alexander S. Ecker, and Matthias Bethge. "Image Style Transfer Using Convolutional Neural Networks". In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2016, pp. 2414–2423. URL: https://arxiv.org/abs/1508.06576 (visited on 04/13/2026).

[2] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E. Hinton. "ImageNet Classification with Deep Convolutional Neural Networks". In: *Advances in Neural Information Processing Systems (NeurIPS)*. Curran Associates, Inc., 2012, pp. 1097–1105. URL: https://papers.nips.cc/paper/2012/hash/c399862d3b9d6b76c8436e924a68c45b-Abstract.html (visited on 04/13/2026).